ASHLEY BLACKWELL

Applied Data Scientist | Machine Learning | Research Data & Visualization | Digital Humanities
Eppstein, Germany • Authorized to Work in Germany • No Sponsorship Required
ashleybrookeblackwell@gmail.com • +49 1522 773 0525 • German: Basic (actively learning)
LinkedIn • GitHub • Hugging Face • Tableau Public

Professional Summary

Applied data scientist with an MS in Data Analytics and BA in History & Art, combining machine learning, model evaluation, data visualization, and 13+ years of operational leadership. Experienced developing and deploying systems across NLP, computer vision, predictive modeling, and network analysis, with strengths in evaluation, explainability, human-in-the-loop methods, and translating complex requirements into usable tools. Interested in interdisciplinary research software and computational approaches to historical, cultural, and human-centered research.

Technical Skills

Programming and Data: Python, SQL, pandas, NumPy, SciPy, Apache Spark, PySpark

Machine Learning and Evaluation: scikit-learn, LightGBM, HistGradientBoosting, Spark MLlib, feature engineering, model calibration, class-imbalance mitigation, SVMs, ensemble modeling, leakage auditing, external validation

Deep Learning and Computer Vision: TensorFlow, Keras, PyTorch, neural networks (including CNNs, RNNs, and GANs/cGANs), EfficientNet, Vision Transformers, transfer learning, Grad-CAM

NLP, Generative AI, and Responsible AI: Hugging Face Transformers, VADER, IBM Watson NLU, IBM watsonx Prompt Lab, zero-shot and few-shot prompting, constrained-output testing, prompt evaluation, IBM AI Fairness 360 (AIF360), fairness assessment, bias mitigation, human-in-the-loop evaluation

Deployment and Visualization: Gradio, Hugging Face Spaces, GitHub, joblib, Tableau, Matplotlib, NetworkX, PyVis

Selected Projects

Longitudinal Predictive Model Suite for Behavioral Healthcare | *(ongoing project)*
Python | LightGBM | NLP | Responsible AI | Model Packaging & Deployment

- Phase I of this project included a synthetic dataset created via python for use in pipeline development and initial testing of models which will predict potential patient no-shows, treatment-lapse, and possible distress/crisis, as well as a custom VADER-based NLP pipeline, defined monitoring requirements for drift, fairness, calibration, versioning, and incident response, and a dashboard interface in preparation for real-world deployment using the Raspberry Pi5
- Phase I achieved 0.892 AUROC and 0.751 F1 for no-show predictions, with 0.848 AUROC during second validation on the Medical Appointment No Shows dataset, created by Joni Hoppen and Aquarela Analytics
- The current Phase II involves constructing a de-identified longitudinal dataset under a data use agreement to support further training, testing, subgroup analysis, and validation before any deployment decisions are made

Lung CT Cancer Classification and Model Deployment
TensorFlow | EfficientNetB0 | Transfer Learning | Grad-CAM | Gradio | Hugging Face Spaces

- Developed and compared a baseline CNN, an AutoKeras architecture search, and an EfficientNetB0 transfer-learning model for four-class lung CT image classification
- Addressed severe class imbalance through custom loss weighting and evaluated model performance using accuracy, macro-F1, and ROC-AUC
- Fine-tuned EfficientNetB0 and integrated Grad-CAM overlays to support visual inspection of model attention within lung regions
- Packaged the trained Keras model and deployed an interactive Gradio application through Hugging Face Spaces, allowing users to upload images and view class probabilities and Grad-CAM visualizations

The Absent Archive: Andean Heritage Metadata Network

Cultural Heritage Metadata | NetworkX | PyVis | Betweenness Centrality | Digital Collections

- Built an interactive network analysis of Andean digital-collection metadata to investigate systemic omission and material continuity within the archival record
- Used Python, NetworkX, PyVis, and betweenness centrality to identify bridge nodes, structural gaps, and patterns of representation across connected metadata records, with particular attention to quipu and Andean material culture

Professional Experience

Strategic Advisor | *2024–Present*
Jones Behavioral Health, Inc.

Augusta, GA • Remote from Germany

- Advise the CEO on corporate strategy, operational compliance, policy development, regulatory planning, and executive decision support
- Maintain organizational compliance frameworks and support the development and implementation of operational and regulatory policies

Chief Operating Officer | *2011–2024*
Jones Behavioral Health, Inc.

Augusta, GA • Behavioral Health Organization • 20+ Clinicians, 10+ Support Staff

- Directed administrative, operational, compliance, technology, and workforce support functions for a multidisciplinary behavioral health organization with 20+ clinicians and 10+ support staff
- Scaled active patient volume from 250 to 430 (72% growth) through operational optimization, improved access to care, and patient-centric workflows
- Designed and deployed operational KPI dashboards supporting executive decision-making, clinician productivity tracking, and revenue performance analysis
- Established data-handling protocols, role-based access governance, and secure remote service workflows within the Kareo EHR infrastructure
- Managed grant writing and funding strategy to secure resources supporting organizational growth and regulatory compliance

Brand Designer, UI/UX | *2018–2019*
Superside (formerly Konsus)

Fully Remote • On-Demand Creative-as-a-Service Platform • 750+ Employees, 72 Countries

- Delivered data visualization, executive presentation design, UI/UX systems, and information design solutions for global enterprise clients in a high-velocity creative services environment
- Translated complex datasets, business concepts, and performance metrics into clear visual narratives and analytics dashboards for executive, client-facing, and cross-functional audiences
- Managed concurrent international project lifecycles, ensuring pixel-perfect creative and analytical design work across strict, variable deadlines

Education

MS, Data Analytics | University of Maryland Global Campus *(4.0 GPA, President's List)* | *2026*
BA, History & Art | Augusta University *(4.0 GPA, Dean's List)* | *2023*

Certifications

Graduate Certificate in Business Analytics | *University of Maryland Global Campus* | *2026*
Deep Learning Specialization | *DeepLearning.AI* | *2026*
Google IT Automation with Python Professional Certificate | *Google, Coursera* | *2026*

Honors & Professional Associations

Upsilon Pi Epsilon (UPE) - International Honor Society for Computing and Information Disciplines • *Member since 2026*
Phi Kappa Phi Honor Society - Multidisciplinary Graduate Academic Honor Society • *Member since 2026*